Politely — Rewriting Impolite Sentences with an Encoder–Decoder Built from Scratch

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1. Introduction

Task. Given a blunt or impatient English sentence, generate a polite sentence with the same content (*"Send me the report now."* → *"Could you please send me the report as soon as possible?"*). It is a sequence-to-sequence paraphrase task: the model learns $P(y \mid x) = \Pi\ P(y_t \mid y_<t, x)$. A good output must be **polite**, **faithful** (who / what / when kept) and **fluent**.

2. Related concepts

An **encoder** (LSTM) reads the input and a **decoder** (LSTM) generates the output token by token. **Bahdanau attention** lets the decoder look at every input state at each step instead of one fixed vector. A **copy mechanism** (pointer-generator) mixes generating a vocabulary word with copying an input word, useful because names, objects and numbers must be carried over.

3. Dataset

Public parallel corpora close to the task (e.g. ParaDetox) are full of insults, so the dataset was built for the project by a reproducible program from hand-written situations, people, objects and phrasings: **9 balanced categories** (request, reminder, refusal, criticism, complaint, disagreement, question, stop, announcement), 12 impolite and 6 polite phrasings per category, no offensive word. Each input has 2 references in train and 3 in evaluation. Splits are made **by content** (0 sentence shared). **Version 1** (`data_v1/`, office life, 1,371-word vocabulary, 10,800 training inputs) contained incoherent or ungrammatical pairs (*"Move your car out of the library"*, *"Call Sarah back. Call it today."*) and gave 0–1 % success on hand-written sentences; **version 2** (`data/`, final) removed them and widened the vocabulary to everyday life.

Split	train	valid	test	test_unseen	challenge
Inputs / pairs	14,400 / 28,800	1,350 / 4,050	1,350 / 4,050	900 / 2,700	72 / 72
Content	—	for choices	seen phrasings	**phrasings never seen**	**written by hand**

4. Preprocessing

Unicode normalisation, lowercasing (case is restored for display), punctuation as separate tokens (*?* vs *.* matters), word tokenisation with contractions kept (*don't*). Vocabulary built **on train only** (words seen ≥ 2 times): **1,805 entries** with `<PAD>` `<SOS>` `<EOS>` `<UNK>`; padding masked, truncation at 40 tokens. Example: *close the window now .* → `390 4 573 58 5 2` → 128-dimensional embeddings.

5. Architecture

Embedding 128 (shared) → **LSTM encoder** (1 layer, hidden 256) → **attention** (score = $v^T \tanh(W_1 h + W_2 s)$, softmax, context vector) → **LSTM decoder** fed [previous word; context] → linear → softmax over 1,805 words, stop at `<EOS>`. **Copy switch** $p_gen = \sigma(w \cdot [c; s; e])$. Models: **A** no attention (1.30 M parameters), **B** + attention (1.66 M), **C** + copy (1.67 M, final), D = GRU, E = hidden 128 / 512, F = teacher forcing 0.5.

6. Training

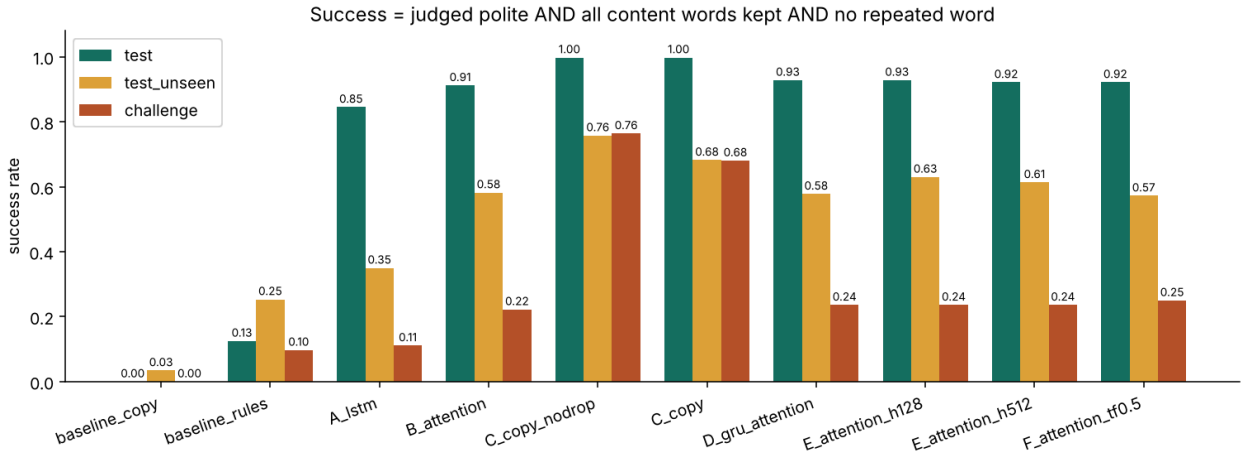
Token cross-entropy (`<PAD>` ignored), Adam (lr 1e-3, halved on plateau), batch 64, dropout 0.3, gradient clipping 1.0, teacher forcing 1.0, early stopping (patience 3, ≤ 30 epochs), CPU (~90 s / epoch). Each model first passed a **sanity check** (memorising 40 pairs: loss 7.5 → 0.07).

7. Experimental results

Success = judged polite by our logistic-regression classifier **and** all content words kept **and** no repetition. Model and decoding (greedy, beam 3/5, sampling, top-k) were chosen on validation: **C + beam 5**.

System	test	test_unseen	challenge	BLEU (test)
Copy baseline / rule-based baseline	0.000 / 0.125	0.034 / 0.253	0.000 / 0.097	47.5 / 48.2
A · LSTM, no attention	0.845	0.350	0.111	74.2
B · + attention	0.912	0.582	0.222	77.2
C · + copy, beam 5 (final)	**0.999**	**0.787**	**0.764**	77.3
C judged by hand: *acceptable*	0.975 (40)	0.375 (40)	0.528 (72)	—

Attention speeds up learning (best epoch 24 → 5) and keeps more content; **copying is the largest gain** (challenge 0.22 → 0.76). Hidden size 128/256/512, GRU and teacher forcing 0.5 give **no gain** (0.91–0.93 on test); word dropout made results worse. No overfitting: validation loss stays at or below training loss.



Success rate of every system on the three test sets.

8. Generated examples

✓ Turn the music down. → Could I ask you to turn the music down? Thank you! · *✓ You're wrong. Water boils at 100 degrees. → Actually, I think water boils at 100 degrees. Could you double-check?* · *✗ I won't lend you any more money. → Would it be possible for you to lend you any more money?* (meaning reversed) · *✗ You were supposed to pay the rent yesterday. → … could you pay the rent yesterday?* (tense).

9. Discussion

The model learns the polite *frames* very well and copies content reliably, but chooses the frame from a few cue words: unseen rude openers are copied as content and grammar is not adapted. The automatic metric agrees with human judgement in-domain but is too optimistic outside it. **Limits:** phrasings from a finite set, bag-of-words politeness classifier, word-level vocabulary. **Improvements:** more varied data, subword units, a Transformer with copy, a larger human evaluation.

10. Conclusion

A small encoder–decoder built from scratch rewrites impolite sentences politely; attention and above all the copy mechanism keep the content (17 % → 53 % fully acceptable on hand-written sentences), while a bigger model does not help and better data does. Lesson: evaluate far from the training data and check metrics by hand.